



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
UNIT TEST
21-02-2021

JEE(Main + Advanced) : NURTURE PHASE-I(B) & NURTURE LIVE PHASE-I(B)

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks : 15)

- This section contains **FIVE** questions.
 - Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
 - For each question, darken the bubble corresponding to the correct option in the ORS.
 - For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases
-

1. If ' ρ ' is the mean density of the earth, R is the radius, g is the acceleration due to gravity at the surface of earth and G is the universal gravitational constant then which equation is dimensionally correct :

(A) $\rho = \frac{3g}{4\pi RG}$

(B) $\rho = \frac{3g^2}{4\pi RG}$

(C) $\rho = \frac{GR}{12\pi g}$

(D) $\rho = \frac{3R}{4\pi gG}$

2. A force of 12 N and another force of 5 N can be applied together to produce the effect of a single force. Which of the following force can not be obtained ?

(A) 14 N

(B) 8 N

(C) 18 N

(D) 16 N

Space for Rough Work

3. In a new system, the unit for mass is α kg, unit of length is β meter, and unit of time is γ second. The value of 1 unit of power in this new system will be :
- (A) $\alpha\beta^2\gamma^{-2}$ Watt (B) $\alpha\beta^3\gamma^{-3}$ Watt (C) $\frac{\gamma^3}{\alpha\beta^2}$ Watt (D) $\alpha\beta^2\gamma^{-3}$ Watt
4. The position of a particle at time t is given by the equation $x(t) = \frac{v_0}{\alpha}(1 - e^{-\alpha t})$ where v_0 is a constant and $(\alpha > 0)$. The dimensions of v_0 and α are respectively :-
- (A) $M^0L^0T^0$ and T^{-1} (B) $M^0L^1T^{-1}$ and T (C) M^0LT^{-1} and T^{-1} (D) M^1L^1T and T^{-2}
5. If instead of mass, length and time as fundamental quantities, we choose velocity, acceleration and force as fundamental quantities and express their dimensions by V , A and F respectively, then the dimensions of Young's modulus 'Y' will be as expressed as : Given $[Y] = [ML^{-1}T^{-2}]$
- (A) $[FA^2V^{-4}]$ (B) $[F^2V^{-1}A]$ (C) $[FA^2V^{-1}]$ (D) $[FAV^{-2}]$

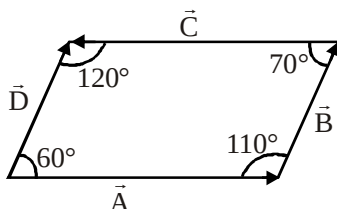
Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
 - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
 - For each question, choose the correct option(s) to answer the question.
 - Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
 - **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.
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6. If A and B are two physical quantities having different dimensions then which of the following operation is meaningless ?
- (A) $A + B$ (B) $A \times B$ (C) $A - B$ (D) $\frac{A+B}{A-B}$
7. If $E = Ae^{-Bx}$ where E is energy and x is length then
- (A) dimension of A is same as energy & B is $\frac{1}{\text{length}}$
- (B) dimension of A is same as power $\times$ time & B is $\frac{1}{\text{Velocity} \times \text{time}}$
- (C) dimension of A is same as ML^2T^{-2} & B is L^{-1}
- (D) None of these

Space for Rough Work

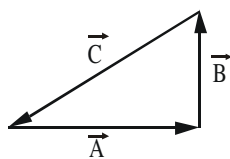
8. Which of the following sets cannot enter the list of fundamental quantities in any newly proposed system of units?
 (A) length, mass and velocity (B) pressure, density and velocity
 (C) force, velocity and time (D) force, momentum and time
9. Which of the following statements is/are **INCORRECT**?
 (A) A dimensionless physical quantity has no unit.
 (B) A physical quantity having no unit is dimensionless.
 (C) A dimensionally incorrect equation must be incorrect.
 (D) Two quantities of same dimensions must represent same physical quantities.
10. When a transverse wave travels in a medium, the displacement (y) of a particle located at x at time t is given by $y = a \sin(bt - cx)$ where a , b and c are constants.
 (A) The dimension of ' a ' is the same as that of ' y '
 (B) The SI unit of b is second
 (C) The dimensional formula of c is $[M^0L^{-1}T^{-1}]$
 (D) The dimension of $\frac{b}{c}$ is same as of velocity.
11. In the given figure



- (A) Angle between $\vec{A}$ and $\vec{B}$ is 110° (B) Angle between $\vec{C}$ and $\vec{D}$ is 60°
 (C) Angle between $\vec{B}$ and $\vec{C}$ is 110° (D) Angle between $\vec{B}$ and $\vec{C}$ is 70°

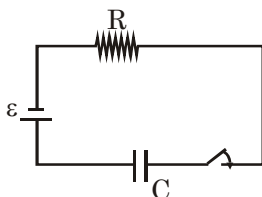
Space for Rough Work

12. Which expression is/are correct concerning the vectors shown in the sketch?



- (A) $\vec{C} = \vec{A} + \vec{B}$ (B) $\vec{C} + \vec{A} = -\vec{B}$ (C) $\vec{A} + \vec{B} + \vec{C} = 0$ (D) $|\vec{C}| = |\vec{A} + \vec{B}|$

13. Current flowing in a RC charging circuit is given by $I = a(1 - e^{-\frac{t}{RC}})$ (I = current, t = time, R = resistance, C = capacitance, $P = I^2R$, P is power)



- (A) SI unit of a is ampere
 (B) Unit of RC is hertz
 (C) $\frac{t}{RC}$ is dimensionless
 (D) Dimension formula of resistance is $M^1L^2T^{-3}I^{-2}$

Space for Rough Work

SECTION-II : Numerical Value Type (Up to second decimal place)

No question will be asked in section II

SECTION-III : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
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Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

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1. If the angle between $\vec{A}$ and $\vec{B}$ is 60° . Then the value of $|\hat{A} + \hat{B}|$ is $\sqrt{x}$ and $|\hat{A} - \hat{B}|$ is $\sqrt{y}$. Find the value of $(x + y)$.
 2. In the given equation $y = A\left(1 - e^{-\frac{yt}{v}}\right)$; t is time and v is velocity. If dimensional formula of y is $M^x L^y T^z$ then value of $(x + 2y + z)$ is equal to
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Space for Rough Work

3. Force F & density ρ are related as $F = \frac{\alpha}{\beta + \rho}$. If dimension of length in α is 'x' and dimension of length in β is 'y' then write the value of $|x + y|$ in OMR sheet.
4. Energy of a particle is 5J. If unit of length and time are doubled and unit of mass is halved then numerical value of the energy in this new system will be 5n. Value of n is.
5. The expression for power in terms of Planck's units is given $h^\alpha C^\beta G^\gamma$ where h is the Planck's constant given by ($h = E/f$) E = energy, f = frequency, C is the speed of light, G is the gravitational constant. The value of $\alpha + \beta + \gamma$ is

Space for Rough Work

PART-2 : CHEMISTRY**SECTION-I(i) : (Maximum Marks : 15)**

- This section contains **FIVE** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
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Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

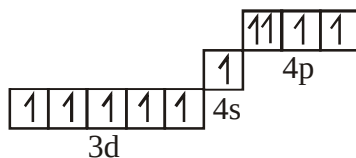
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1. x^- , y^{-2} and z^{-3} are isotonic and isoelectronic. Thus increasing order of atomic number of x, y and z is
(A) $x < y < z$ (B) $z < y < x$ (C) $x = y = z$ (D) $z < x < y$
 2. Which of the following sets of quantum numbers represents the highest energy of an electron in multi electron system?
(A) $n = 4, l = 0, m = 0, s = +1/2$ (B) $n = 4, l = 2, m = 1, s = +1/2$
(C) $n = 5, l = 0, m = 1, s = -1/2$ (D) $n = 4, l = 1, m = 0, s = -1/2$
 3. Among the following pair of orbitals which orbits will experience the larger effective nuclear charge
(i) 2s and 3s (ii) 4d and 4f (iii) 3d and 3p
(A) 2s, 4d and 3p (B) 3s, 4f and 3p (C) 2s, 4f and 3d (D) 2s, 4d and 3d
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Space for Rough Work

4. Given this set of quantum number for a multi electronic atom $2, 0, 0, \frac{1}{2}$ and $2, 0, 0, -\frac{1}{2}$.

What is the next higher energetic allowed set of n and l quantum numbers

- (A) $n = 2, l = 0$ (B) $n = 2, l = 1$ (C) $n = 3, l = 0$ (D) $n = 3, l = 1$
5. In the following electronic configuration, which rules are violated.



- (I) Hund's rule (II) Pauli's exclusion principle (III) Aufbau principle
- (A) I and II (B) I and III (C) II and III (D) I, II and III

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
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 - **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.
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6. Which of the following order of ionic radius is/are **CORRECT** ?
(A) $\text{Na}^+ < \text{K}^+$ (B) $\text{Ca}^{+2} < \text{K}^+$ (C) $\text{S}^{2-} > \text{O}^{2-}$ (D) $\text{H}^- > \text{F}^-$
7. Which of the following statement(s) is/are **CORRECT** ?
(A) The possible values of all the four quantum numbers for the 19th electron of Cr is $n = 4$, $l = 0$, $m = 0$ and $s = +1/2$
(B) The possible values of all the four quantum numbers for the 21st electron of Sc is $n = 3$, $l = 2$, $m = -2$ and $s = +1/2$
(C) The possible values of all the four quantum numbers for the 7th electron of N is $n = 2$, $l = -1$, $m = 1$, $s = -1/2$
(D) The possible values of all the four quantum numbers for the electron of H is $n = 0$, $l = 0$, $m = 0$, $s = -1/2$
8. Which of the following paramagnetic ion have same number of unpaired electrons.
(A) Fe^{+3} (B) Mn^{+2} (C) Co^{+4} (D) Cr^+
-

Space for Rough Work

9. Which of the following statement is **CORRECT** for d_{z^2} orbital.
- (A) It has value of $l = 2$ (B) It has four lobes
(C) It has two nodal cone (D) It is axial orbital
10. Which of the following statement(s) is/are **INCORRECT** regarding to Lothar Meyer curve (atomic wt v/s atomic volume)
- (A) Alkali metals are present on ascending position of the curve
(B) Halogen are present on descending position of the curve
(C) Alkaline earth metals are present at the peak of the curve
(D) Alkali metals are present at the peak of the curve.
11. The radius of which of the following species is/are less than that of the radius of O^{2-} ?
- (A) F^- (B) O^- (C) Se^{2-} (D) Na^+
12. Which of the following statement(s) regarding modern periodic table is/are **CORRECT**?
- (A) Ar is considered as a bridge element.
(B) Electronic configuration of La(57) is $(Xe)6s^25d^1$.
(C) 13th group follows irregular trend of atomic radius.
(D) 15th group follows regular trend of atomic radii.
13. Which of the following statement(s) is/are **CORRECT** ?
- (A) In a p-orbital the number of nodal planes equal to the azimuthal quantum number
(B) The electronic energy of the H-atom depends only on n and is independent of l and m
(C) Those atoms in which all electrons are paired slightly repelled by an external magnetic field and are said to be paramagnetic
(D) Those atoms in which one or more electrons have unpaired spins are strongly attracted to magnetic field and are said to be diamagnetic

Space for Rough Work

SECTION-II : Numerical Value Type (Up to second decimal place)

No question will be asked in section II

SECTION-III : (Maximum Marks : 15)

- This section contains **FIVE** questions.
 - The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
 - For each question, darken the bubble corresponding to the correct integer in the ORS.
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Zero Marks : 0 In all other cases.
-

1. Find the number of orbitals, which have atleast one common nodal plane with d_{zx} .
 $p_x, p_y, p_z, d_{xy}, d_{yz}, d_{z^2}, s, d_{x^2-y^2}$
 2. Find the difference in number of electrons of s-orbitals in Fe and Ni^{+2} .
 3. Find the value of x, if Mn^{+x} has magnetic moment value is 3.87 B.M.
-

Space for Rough Work

4. Find the total number of **CORRECT** orders for the property indicated against them
- (i) $P > S > F$ (atomic radius)
 - (ii) $Na > S > N$ (total number of electron in outermost shell)
 - (iii) $Na < Mg < Ca$ (Z_{eff} value on last electron)
 - (iv) s-block < d-block < p-block (total number of non-metal elements)
5. Find the number of cation(s) which is/are diamagnetic.
 Na^+ , Fe^{+2} , Fe^{+3} , Cu^{+2} , Ni^{+2} , Cl^- , S^{2-}
-

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks : 15)**

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- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
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Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. The number of solutions of the equation $x^2 - 4x + 4 + |x - 2| = 0$ is -

- (A) 1 (B) 2 (C) 3 (D) 4

2. If $\frac{(\sqrt{x} - \sqrt{y})^2 + 4\sqrt{xy}}{x - y} = \frac{7}{3}$ ($x, y \geq 0, x \neq y$), then $x : y$

- (A) 5 : 2 (B) 2 : 5 (C) 25 : 4 (D) 4 : 25

Space for Rough Work

3. If A, B and C are three non disjoint sets then $A - (B \cap C) =$
- (A) $(A - B) \cup (A - C)$ (B) $(A - B) \cap (A - C)$
(C) $(A \cup B) - (A \cap C)$ (D) $(A - B) \cap C$
4. In a survey of 200 students of a school, it was found that 120 study Mathematics, 90 study Physics and 70 study Chemistry, 40 study Mathematics and Physics, 30 study Physics and Chemistry, 50 study Chemistry and Mathematics and 20 none of these subjects, then the number of students who study all the three subjects is :
- (A) 20 (B) 30 (C) 40 (D) 50
5. The number of positive integer pairs satisfying $x^2 - y^2 = 2010$ is -
- (A) 0 (B) 2 (C) 5 (D) more than 5
-

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

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6. $\frac{1}{\sqrt{1}+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} + \dots + \frac{1}{\sqrt{8}+\sqrt{9}}$ is a -

- (A) rational number (B) integer
(C) irrational number (D) prime number

7. If A and B are two sets and U is the universal set such that $n(U) = 700$, $n(A) = 200$, $n(B) = 300$ and $n(A \cap B) = 100$, then $n(A' \cap B')$ is not equal to

- (A) 400 (B) 300 (C) 500 (D) 600

Space for Rough Work

8. If $A = \{1, 2, 3, \{5, 7\}, 8\}$ then correct statement is

- (A) $\{5\} \in A$ (B) $\{5, 7\} \in A$ (C) $\{1, 2, 3\} \in A$ (D) $\{\{5, 7\}\} \subset A$

9. Let A, B, C be real number such that $A + B + C = 7$ and $\frac{1}{A+B} + \frac{1}{B+C} + \frac{1}{C+A} = \frac{7}{10}$ then

$\frac{A}{B+C} + \frac{B}{C+A} + \frac{C}{A+B}$ is not equal to -

- (A) $\frac{19}{10}$ (B) $\frac{17}{10}$ (C) $\frac{9}{7}$ (D) $\frac{10}{7}$

10. If $\sqrt{16 - 2\sqrt{33} - 2\sqrt{6} + 2\sqrt{22}} = \sqrt{a} - \sqrt{b} + \sqrt{c}$ then which of the following can be true

- (A) $a = 11, b = 3, c = 2$ (B) $a = 11, b = 2, c = 3$
(C) $a = 2, b = 3, c = 11$ (D) $a = 2, b = 11, c = 3$

Space for Rough Work

11. If $a + b + c = 3$, $a^2 + b^2 + c^2 = 13$, $a^3 + b^3 + c^3 = 27$, then the value of $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$ is not equal to
- (A) $\frac{1}{3}$ (B) 3 (C) 1 (D) $\frac{1}{9}$
12. The sum of all real values of x satisfying the equation $(x^2 - 5x + 5)^{x^2 + 4x - 60} = 1$ is not equal to
- (A) 5 (B) 3 (C) -4 (D) 6
13. If a polynomial $ax^3 + bx^2 + cx + d$ vanishes at $x = 1$ and at $x = -2$ and leaves the remainder 4 & 28 when divided by $x + 1$ and $x - 2$ respectively, then d is not equal to
- (A) 14 (B) 7 (C) -7 (D) -2

Space for Rough Work

SECTION-III : (Maximum Marks : 15)

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Zero Marks : 0 In all other cases.
-

1. The number of possible ordered pairs (x,y) (where x and y are integers) which satisfy $\frac{3}{x} + \frac{5}{y} = \frac{21}{xy} + 1$ is
 2. If a679b is a five digit number that is divisible by 72, then a + b is equal to
 3. If P(x) is a polynomial of degree 3 such that the coefficient of x^3 is $-\frac{1}{4}$. If P(x) is divided by (x-1), (x-2) & (x-3), remainders are 1,2 & 3 respectively. The remainder when P(x) is divided by (x + 1) is
-

Space for Rough Work

4. If $x = \sqrt[3]{7+5\sqrt{2}} - \frac{1}{\sqrt[3]{7+5\sqrt{2}}}$, then value of $x^3 + 3x - 14$ is equal to _____.
5. If $\sqrt{11-2\sqrt{30}} - \sqrt{10-4\sqrt{6}} = \alpha - \sqrt{\beta}$, where α, β are integers & β is not a perfect square then $\alpha + \beta$ is equal to _____.

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	5	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	15
I(ii)	One or more correct option(s)	8	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–2 In all other cases	32
III	Single digit Integer (0-9)	5	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	15

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator